

Digital Platform for Social Welfare Services Management in Sri Lanka



Group ID : 25 - 26J - 320
CoEAI • Information Technology

Our Team

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Research Problem & STAKEHOLDERS

Sri Lanka's welfare system (Samurdhi, disability, elderly pensions) suffers from manual processes, fragmented data across 6+ departments, language barriers, and approval delays causing eligible citizens to miss aid while ineligible ones receive it.

60%

Language
Accessibility
Barriers

25%

Benefits
Misallocated
or Duplicated

>40%

Applications
Severely
Delayed

STAKEHOLDERS & IMPACT



Citizens

Eligible welfare applicants facing language barriers, long delays and no healthcare guidance



Divisional Officers

Officers processing applications manually without AI tools, dashboards or real-time data



Policymakers

Decision-makers lacking AI insights, fraud detection, geographic risk maps and analytics

Research GAP & Objectives

Eligibility Prediction & Healthcare Recommendation

Predict who qualifies for welfare programs and suggest hospitals or doctors (e.g., for pregnant mothers or elderly patients).

Multilingual Conversational Assistance

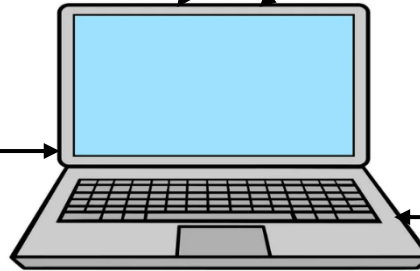
Build a chatbot in Sinhala, Tamil, and English to guide users on applications and documents (e.g., check application status or required forms).

AI-GIS Welfare Insights Platform

Map areas with high welfare demand to help policymakers (e.g., regions with many chronic patients or low-income families).

Investment Plan Recommendation

Recommend financial support or investment options to users (e.g., microloans or savings plans).



Feature	Current Sri Lanka (Manual System)	Proposed AI Platform
AI-Based Eligibility Prediction	✗ None	✓ GB · RF · DT · LR (91.80%)
Healthcare Recommendation	✗ None	✓ Profile-based ranking engine
Multilingual Chatbot Support	✗ None	✓ Sinhala · Tamil · English (NLP)
AI Financial Guidance	✗ None	✓ K-Means + Collaborative Filtering
GIS Spatial Risk Mapping	✗ None	✓ AI + GIS hotspot prediction

SOLUTION OVERVIEW — WHAT WE BUILT

AI-Powered Digital Welfare Platform

01



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**Eligibility Prediction
& Healthcare Rec.**

Gradient Boosting
Random Forest
Decision Tree
Logistic Regression
1,830 records · 91.80% accuracy

02



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Shamika A.G

**Multilingual Chatbot
(Sinhala/Tamil/English)**

NLP · Intent Recognition
Named Entity Recognition
mBERT · Rasa · spaCy

03



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Dissanayake D.M.V.S

**AI Investment Plan
Recommendation Module**

K-Means Clustering
Collaborative Filtering
Progress Tracking

04



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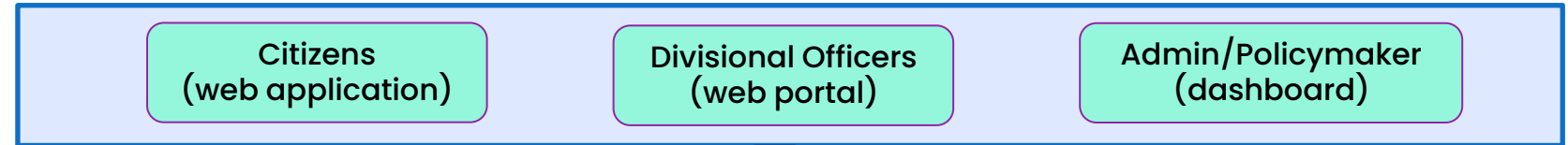
Wickramasinghe R.P.Y.N

**AI + GIS Vulnerable
Population Mapping**

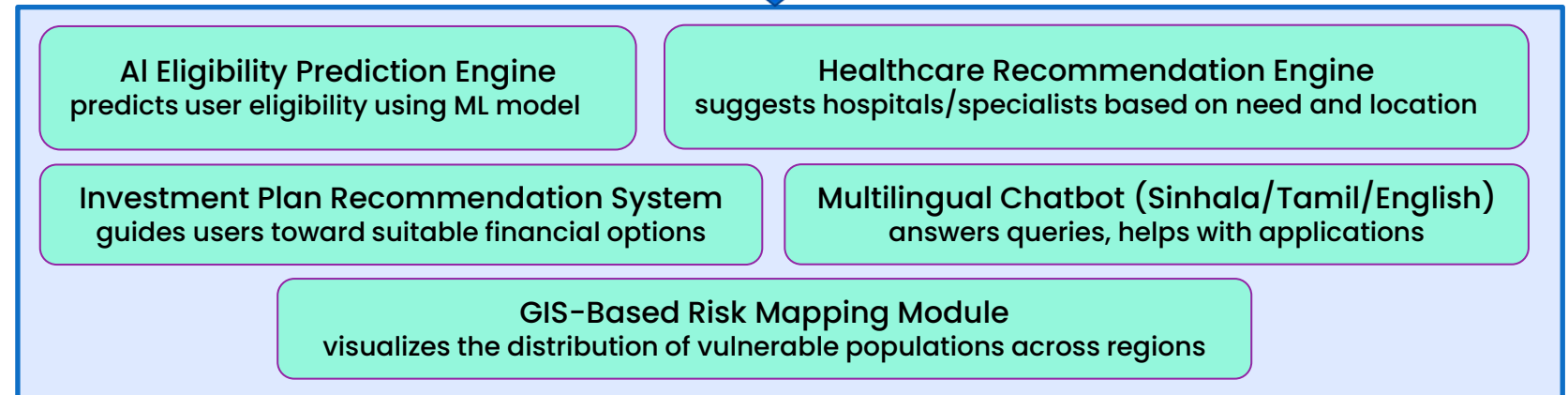
Spatial Clustering · Heatmaps
Predictive Analytics
GIS Dashboards

Overall System Diagram

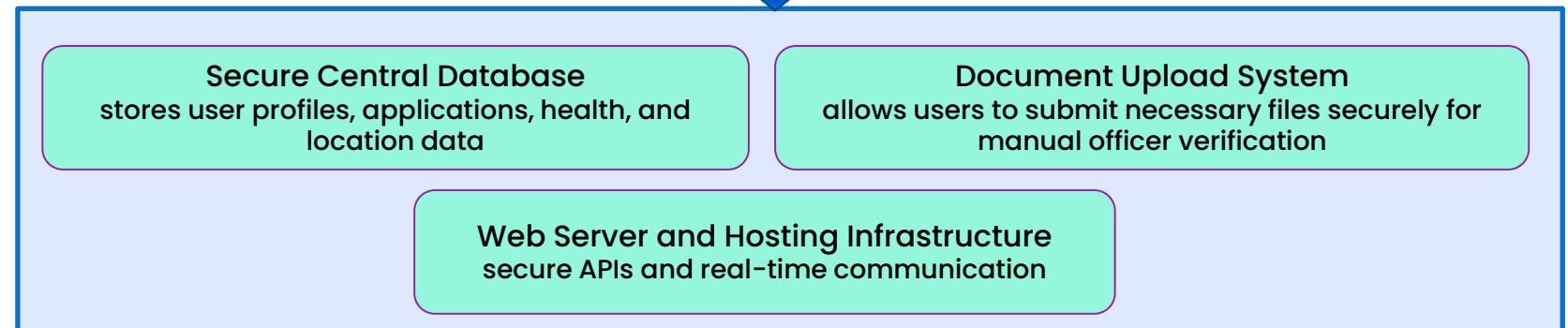
User Interface Layer



AI Intelligence Layer



Infrastructure Layer



Commercialization & Impact

Target Market:

Governments, healthcare providers, NGOs (Main client)

- Position as a national welfare platform for the Sri Lankan government.
- Provide specialized modules for NGOs to manage targeted welfare programs.

Researchers & Private Companies (Data Access)

- Offer secure, anonymized datasets to universities, health researchers, and private sector organizations.
- Support evidence-based policymaking, academic studies, and market analytics while ensuring data privacy.

Advertisement Revenue on Platform

- Allow ethical, welfare-related advertisements (e.g., healthcare services, educational programs, insurance).
- Generate continuous revenue without affecting citizen services.

REVENUE STREAMS

Government Licensing	Annual SaaS license to DSD/Ministry
Institutional B2B	Hospital & clinic integration fees
Premium Tier (D2C)	Advanced AI reports for individuals
Data Insights (Ethical)	Anonymized research datasets

ESTIMATED BUDGET

Cloud VPS Hosting	LKR 10,000/mo
App Store (iOS)	LKR 7,754/yr
Play Store (Google)	LKR 30,700 once
Domain + SSL	LKR 5,000/yr
3rd Party APIs	LKR 15,000/mo
Marketing Launch	LKR 50,000 once
Maintenance	LKR 25,000/mo

DEMO PLAN – 30 MINUTE LIVE DEMONSTRATION

01

Manawadu G.M.S.T (IT22926494)

Eligibility Prediction & Healthcare Recommendation

1. Open Streamlit app — show 1,830-record dataset overview
2. Enter citizen profile → AI prediction: ELIGIBLE / NOT ELIGIBLE
3. Healthcare: specialist + hospital + Google Maps + 8 health tips

03

Dissanayake D.M.V.S (IT22366818)

AI Investment Plan Recommendation Module

1. Input low-income user financial profile → K-Means cluster assignment
2. Display personalized micro-investment plan recommendations
3. Show progress tracking dashboard + adaptive follow-up scheduling

02

Shamika A.G (IT22585080)

Multilingual Chatbot (Sinhala / Tamil / English)

1. Submit welfare query in English then Sinhala — show NLP processing
2. Demonstrate intent recognition + entity extraction output
3. Show multi-turn dialogue management + chatbot analytics

04

Wickramasinghe R.P.Y.N (IT22214584)

AI + GIS Vulnerable Population Mapping

1. Open GIS dashboard — show Sri Lanka province heatmap
2. AI model predicting future welfare demand by district
3. Interactive map: filter by condition / age / district + policy insights



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